

Employee Attrition Prediction Using Machine Learning

Project Summary

Objective

The objective of this project was to develop a machine learning system that could forecast an employee's likelihood of leaving a company based on variables like job happiness, monthly income, work-life balance, years of employment, overtime status, and performance-related characteristics. In order for an HR team to apply these insights for retention planning, the project's goals were to compare the efficacy of several classification algorithms on this task and to pinpoint the precise elements that most significantly affect employee turnover.

Dataset Description

For this study, the IBM HR Analytics Employee Attrition dataset was utilized. It includes data on 1,470 workers in 35 different categories, including department, job role, marital status, overtime status, and numerical variables like age, monthly income, and years at the company. Whether an employee left the company (Yes) or stayed (No) is indicated by the target variable, attrition. An initial investigation revealed an attrition rate of 16.12%, indicating that the dataset is unbalanced, with much more workers remaining than departing. This is a crucial factor to take into account when developing and assessing models.

Data Preprocessing

When the dataset was examined for missing values, none were discovered. EmployeeNumber, Over18, StandardHours, and EmployeeCount were the four columns with no predictive value that were eliminated since they were found to be constant or merely identifying data. Binary 1/0 was used to replace the Yes/No format in the target column. One-hot encoding was used to transform categorical features like Department, Job Role, Marital Status, Business Travel, OverTime, and Education Field into numerical form. StandardScaler was then used to standardize the numerical features so that each feature contributed proportionately during model training. Stratified sampling was then used to divide the dataset into training and testing sets in an 80:20 ratio while maintaining the initial attrition ratio in each set.

Machine Learning Models

Three classification algorithms—Random Forest Classifier, Gradient Boosting Classifier, and Logistic Regression—were put into practice and contrasted. Because the dataset is unbalanced, class weighting was used during training to prevent the majority class from overpowering the minority class of departing employees. This was done by using the `class_weight='balanced'` parameter for Random Forest and Logistic Regression and balanced sample weights for Gradient Boosting, which does not directly support this parameter. Since accuracy can be deceptive on an unbalanced dataset like this one, the model's performance was assessed using Precision, Recall, F1-Score, ROC-AUC Score, and Confusion Matrix.

Data Visualization

Several visualizations were made in order to gain a better understanding of the dataset and the models that were produced, such as:

Attrition rate by occupational role and department

Monthly income box plot for departing and remaining employees

Heatmap of confusion matrix for the top-performing model

The top ten factors influencing attrition are displayed in a horizontal bar chart.

ROC curve that contrasts the three models

These visualizations made it possible to identify which factors, and which employee groups, carried the highest risk of attrition.

Key Findings

The data showed that attrition risk is concentrated rather than evenly spread across the company. Sales Representatives had the highest attrition rate of any role, at nearly 40% — more than double the next-highest role. Employees who worked overtime left at roughly three times the rate of those who did not (about 30% versus 10%), and employees in their first year at the company left at a rate of 34.5%, far higher than any other tenure group. Monthly income was also linked to attrition, with employees who left earning noticeably less on average than those who stayed, although income alone did not explain the pattern as strongly as overtime and role-specific factors did. Among the three models tested, Logistic Regression achieved the best ROC-AUC score (0.798) and, more importantly, the highest recall (61.7%) — meaning it identified far more of the employees who actually left than either Random Forest or Gradient Boosting, both of which achieved higher precision but missed the majority of true leavers.

Conclusion

This project demonstrated how machine learning can be used to identify employees at risk of leaving and to surface the specific factors driving attrition, rather than relying on guesswork. Through data preprocessing, exploratory analysis, model building and careful evaluation, the project produced clear, numbers-backed insights — particularly around overtime, job role and early tenure — that an HR team could act on directly. The model also has limitations worth noting: it is trained on a single company's data and a relatively small number of historical leavers, so it should be used as a risk indicator to prompt a conversation, not as a guaranteed prediction. Future work could explore a larger dataset, techniques such as SMOTE for handling class imbalance, and more advanced algorithms such as XGBoost to further improve predictive performance.

Prepared By: Vaishnav Kurade

Project: Employee Attrition Prediction Using Machine Learning

Tools Used: Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Google colab